Worksheet 7.1 — Planning Your Mathematical Exploration

Mathematics: Analysis & Approaches SL (IB) · Unit 7: The Exploration (Internal Assessment)

The Big Ideas

Use this worksheet to plan your Exploration before you start writing — a clear plan up front saves far more time than it costs.

- Five criteria: A Presentation, B Mathematical communication, C Personal engagement, D Reflection, E Use of mathematics.
- A strong aim names a specific, answerable question — not just a general area of interest.
- Real (self-collected or public) data strengthens both Criterion C and E far more than invented numbers.

Worked Examples

Example 1 — Modelling rumour spread

Model how news spreads through a school using geometric-growth ideas, compared against a small survey you run yourself.

Solution. **Why it works:** uses Unit 1 sequences meaningfully, and pairs a model with real, self-collected data.

Example 2 — Optimizing a drink can

Use Unit 5's optimization techniques to find a cylinder's minimal-material dimensions for a fixed volume, then compare to real cans.

Solution. **Why it works:** a genuine calculus application with a natural extension — comparing theory to reality invites real reflection.

Example 3 — Correlation in an unexpected pair

Investigate whether two seemingly unrelated variables show meaningful correlation, using Unit 4's tools on a real dataset.

Solution. **Why it works:** forces careful thought about correlation vs causation — exactly what Criterion D rewards.

Example 4 — Combining two strands

Model a Ferris wheel rider's height with a sinusoidal function (Unit 3), then integrate (Unit 6) to find total vertical distance travelled.

Solution. **Why it works:** combining two strands into one coherent investigation pushes Criterion E from sufficient to excellent, without needing content beyond SL.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Write a one-sentence aim for a topic of your own. Does it name a specific, answerable question?

Question 2

List the specific mathematical techniques from this course your topic will require.

Question 3

Outline your four main sections (introduction/aim, background, mathematical content, conclusion) in one sentence each.

Question 4

Will you use real data? If so, where will it come from? If not, explain why an invented scenario still serves your aim.

Question 5

Identify one way you could extend your basic approach to demonstrate more sophisticated mathematics.

Question 6

What is the biggest risk to your topic being too broad or too narrow? How will you keep it appropriately scoped?

Question 7

If your results don't match your model's predictions, what would you write in your reflection section?

Question Recap & Answer Key

Question Recap

1. Write a one-sentence aim for a topic of your own. Does it name a specific, answerable question?
2. List the specific mathematical techniques from this course your topic will require.
3. Outline your four main sections (introduction/aim, background, mathematical content, conclusion) in one sentence each.
4. Will you use real data? If so, where will it come from? If not, explain why an invented scenario still serves your aim.
5. Identify one way you could extend your basic approach to demonstrate more sophisticated mathematics.
6. What is the biggest risk to your topic being too broad or too narrow? How will you keep it appropriately scoped?
7. If your results don't match your model's predictions, what would you write in your reflection section?

✓ Answer Key

1. A strong aim reads like "Does X follow model Y, and how well?" — specific enough that a stranger could tell exactly what you're investigating
2. Answers will vary; check that every technique listed is one you've actually learned and can justify using
3. Answers will vary; each section should have a clear, distinct purpose — avoid overlap between background and mathematical content
4. Real data (self-collected or public) generally strengthens Criteria C and E; an invented scenario can still work if it's used to illustrate a technique clearly and honestly labelled as illustrative
5. E.g., comparing your model against an alternative one, or testing its reliability outside the original data range
6. Too broad: you won't have room to go deep. Too narrow: there's nothing left to investigate. Aim for one clear question with room for a genuine "extension" step
7. A mismatch is a strength, not a failure — discuss why the model might fail (a missing variable, an unrealistic assumption, measurement error) to show critical reflection